NVIDIA DEEPSTREAM

Crowd Detection & ROI-Based Access Restriction System

Deployed on NVIDIA Jetson Orin | Technical Architecture & Implementation Proposal

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Executive Summary

This document presents the technical architecture and implementation plan for an intelligent Crowd Detection and Region of Interest (ROI) Access Restriction System, built on NVIDIA's DeepStream SDK and deployed on the NVIDIA Jetson Orin edge computing platform.

The system ingests live camera feeds, runs real-time crowd density analysis using NVIDIA's pre-trained people-detection and crowd-counting models, dynamically defines and enforces ROI zones, and triggers automated alerts when individuals attempt to enter restricted areas. The entire inference pipeline runs at the edge — on the Jetson Orin — ensuring low latency, offline capability, and data privacy.

Primary use cases include: crowd detection and real-time monitoring in political campaigns and rallies for safety, turnout analysis, and resource planning.

System Architecture Overview

The pipeline is composed of five integrated layers running on Jetson Orin, orchestrated by the DeepStream SDK:

#	Layer	Function	Key Technology
1	Video Ingestion Layer	Multi-camera RTSP/USB feed ingestion, decode & pre-process	DeepStream NvStreammux, NVDEC Hardware Decoder
2	Inference Layer	People detection, tracking, crowd density estimation per frame	NVIDIA PeopleNet, CrowdCount, TAO Toolkit models
3	ROI Management Layer	Define, load, and enforce polygonal ROI zones on video frames	DeepStream NvDsROI, custom ROI config (.txt / REST API)
4	Violation Detection Layer	Detect when tracked persons enter restricted ROI zones	Custom GStreamer plugin + coordinate boundary logic
5	Alert & Response Layer	Trigger alarms, overlay annotations, send webhook notifications	DeepStream OSD, MQTT/HTTP alerts, onboard buzzer/relay
6	Visualisation & Logging Layer	Display annotated stream, log events, export to dashboard	RTSP re-stream, InfluxDB/Grafana, SQLite local log

Hardware — NVIDIA Jetson Orin

The NVIDIA Jetson Orin is the recommended edge deployment platform for this system. Its dedicated GPU, deep learning accelerators (DLA), and hardware video decode engines make it ideal for running multiple concurrent camera streams with real-time inference.

Jetson Orin Module Comparison

Module	AI Performance	GPU	CPU	RAM	Recommended For	Price (Approx.)
Orin NX 8GB	10 TOPS	1024-core Ampere	6-core Arm Cortex	8 GB	1–4 cameras, light workload	~$299
Orin NX 16GB	20 TOPS	1024-core Ampere	8-core Arm Cortex	16 GB	4–8 cameras, moderate workload	~$399
Orin AGX 32GB	200 TOPS	2048-core Ampere	12-core Arm Cortex	32 GB	8–16 cameras, heavy inference	~$999
Orin AGX 64GB	275 TOPS	2048-core Ampere	12-core Arm Cortex	64 GB	16+ cameras, multi-model pipeline	~$1,499
Orin Industrial	50 TOPS	1024-core Ampere	8-core Arm Cortex	32 GB	Harsh environment / outdoor deploy	~$799

✅Recommendation

For a typical deployment of 4–12 cameras with real-time crowd detection + ROI enforcement, the Jetson AGX Orin 32GB is recommended.

It provides 200 TOPS of AI performance, supports hardware-accelerated decoding of 16 concurrent 1080p streams, and runs the full DeepStream pipeline without frame-drop at 30 FPS per stream.

For cost-sensitive single-site deployments (1–4 cameras), the Orin NX 16GB is a viable lower-cost alternative.

Supporting Hardware Components

Component	Specification	Purpose	Est. Unit Cost
IP Cameras (PTZ or Fixed)	1080p or 4K, RTSP-capable, H.264/H.265	Primary video feed input	$80–$400 each
PoE Network Switch	8–16 port, 802.3af/at PoE	Power & data for IP cameras	$60–$250
NVMe SSD (512GB–1TB)	M.2 NVMe, e.g. Samsung 980 Pro	OS, model weights, local event logging	$60–$120
Active Cooling Kit	Heatsink + fan (Jetson Orin AGX kit)	Thermal management for edge deploy	$20–$40

Relay / Buzzer Module	5V GPIO-controlled relay board	Physical access alert / barrier control	$5–$20
UPS / Power Backup	12V UPS with 2–4 hr runtime	Uptime during power outage	$80–$200
Industrial Enclosure	IP65-rated outdoor enclosure	Weatherproof housing for outdoor sites	$50–$150

NVIDIA DeepStream Pipeline Architecture

NVIDIA DeepStream SDK provides a GStreamer-based pipeline for building high-performance video analytics applications. Each stage of the pipeline is implemented as a GStreamer element (plugin), hardware-accelerated via CUDA and the Jetson Orin's dedicated video processing engines.

Full GStreamer Pipeline Flow

DeepStream GStreamer Element Chain
1. [source_bin] → URI source (RTSP/file/USB) per camera via uridecodebin
2. [nvstreammux] → Batch multiple streams into single buffer (NvBufSurface)
3. [nvinfer] → Primary inference: PeopleNet detector (TensorRT engine, FP16/INT8)
4. [nvtracker] → NvDCF / DeepSORT multi-object tracker — assigns unique Person IDs
5. [nvinfer #2] → Secondary inference: Crowd density / attribute classification (optional)
6. [nvdsanalytics]→ ROI polygon definition, line crossing detection, direction detection
7. [roi_plugin] → Custom GStreamer plugin: checks person bbox centroid vs ROI polygons
8. [nvosd] → On-screen display: bounding boxes, ROI overlays, alert banners
9. [alert_plugin]→ Trigger MQTT/HTTP alert, GPIO relay, event log on violation
10.[nvvideoconvert + nvv4l2h265enc] → Re-encode annotated stream for RTSP output
11.[rtsp_sink] → Output annotated stream to RTSP server for dashboard / NVR viewing

Detailed Pipeline Stages

STEP 1

Multi-Camera Video Ingestion

Multiple IP camera streams (RTSP), USB webcams, or recorded video files are ingested simultaneously. The NvStreamMux element batches all streams into a single NvBufSurface for efficient GPU batch inference.

- Supports up to 16 simultaneous 1080p streams on Jetson AGX Orin
- NVDEC hardware decoder handles H.264 / H.265 decode with zero CPU load
- Configurable input resolution, FPS, and batch size per stream
- Supports RTSP, V4L2 (USB/CSI), file URI, and GigE Vision cameras

STEP

People Detection — NVIDIA PeopleNet

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PeopleNet is an NVIDIA TAO pre-trained model optimised for detecting people in crowded scenes. It runs as a TensorRT-optimised engine on the Jetson Orin GPU, delivering real-time bounding box detections for every person in each frame.

- **NVIDIA PeopleNet v2.6 (NVIDIA NGC catalogue, free to download):** Model
- **ResNet-34 backbone with SSD detection head:** Architecture
- **Person, Bag, Face (configurable):** Classes detected
- **960×544 (default), configurable per deployment:** Input resolution
- **FP16 or INT8 (TensorRT-optimised on Jetson Orin):** Precision
- **~30 FPS per stream at 1080p on Jetson AGX Orin:** Performance
- **Configurable (default: 0.6 for person class):** Confidence threshold

Model Source

PeopleNet is available for free on NVIDIA NGC:
https://catalog.ngc.nvidia.com/orgs/nvidia/teams/tao/models/peoplenet

Download the .etlt model, convert to TensorRT engine using tao-converter on the Jetson Orin.

Pre-built DeepStream config files are included in the NGC download package.

STEP **3**

Multi-Object Tracking

Each detected person is assigned a unique tracking ID that persists across frames. This enables the system to monitor individual movement paths, determine entry into ROI zones, and maintain per-person violation history.

- **NvDCF (NVIDIA Discriminative Correlation Filter) — GPU-accelerated, low latency:** Primary tracker
- **DeepSORT with Re-ID embeddings for higher accuracy in dense crowds:** Alternative
- **Bounding box (x, y, w, h), centroid, unique track ID, class label, confidence:** Output per person
- **Optional OSNet / NVIDIA Re-identification model for handling occlusions:** Re-ID model
- **Configurable max age (frames before track is dropped), min hits to confirm:** Track lifecycle

STEP **4**

Crowd Density Analysis

Beyond person detection, the system uses a crowd density estimation model to compute crowd density heatmaps and head counts per ROI zone. This enables proactive density-based alerts before physical violations occur.

- **NVIDIA CrowdCount (DmCount / CSRNet architecture, available on NGC):** Model
- **Crowd density heatmap (per pixel density) + aggregated count per defined zone:** Output
- **Configurable per ROI: e.g., alert if > 30 people/m² or > 50 people in zone:** Thresholds
- **Density map updated every N frames (configurable for performance tuning):** Update rate

Density Levels & Actions

GREEN (Low density) — < 1 person/m² — No action required
YELLOW (Medium density)— 1–3 persons/m² — Log event, display density overlay ORANGE (High density) — 3–6 persons/m² — Send advisory alert to operator
RED (Critical) — > 6 persons/m² — Trigger alarm, close entry, notify emergency team

STEP **5**

ROI Zone Definition & Management

Regions of Interest (ROIs) are polygonal zones drawn onto the camera's coordinate space. Each ROI is classified as either RESTRICTED (no entry) or MONITORED (density tracking only). The ROI engine checks every tracked person's position against all active ROIs in real time.

ROI Types

ROI Type	Description	Trigger Condition	Action on Trigger
RESTRICTED ZONE	Hard no-entry polygon (e.g., machinery area, VIP lounge, runway)	Any person centroid enters polygon	Immediate alarm + log + alert
BUFFER ZONE	Warning perimeter outside a restricted zone	Person centroid within buffer boundary	Warning alert + operator notification
CROWD LIMIT ZONE	Area with maximum occupancy cap	Person count in zone exceeds threshold	Alert + optional entry gate control
DIRECTION ZONE	One-way corridor or flow control area	Person moves in disallowed direction	Alert + on-screen warning overlay
DWELL ZONE	Area where stationary presence is flagged	Person remains in zone > N seconds	Alert with duration in message
EXCLUSION SCHEDULE	Zone restricted only during defined hours	Person in zone during restricted time window	Alert with time-based enforcement

ROI Configuration Format

ROI zones are defined in a DeepStream analytics configuration file (config_nvdsanalytics.txt). Each zone is specified as a list of polygon vertices in pixel coordinates:

Sample ROI Config (config_nvdsanalytics.txt)

```
[roi-filtering-stream-0]
enable = 1
roi-RF-0 = 120;200;350;200;350;480;120;480   # Restricted Zone 1 (rectangle)
roi-RF-1 = 600;150;900;150;950;400;550;400   # Restricted Zone 2 (polygon)

[overcrowding-stream-0]
enable = 1
```

```
roi-OC-0 = 0;0;1920;0;1920;1080;0;1080      # Full frame crowd monitor
overcrowding-threshold = 50                 # Alert if > 50 persons

[line-crossing-stream-0]
enable = 1
extended = 0
line-crossing-Entry = 960;0;960;1080        # Vertical mid-line crossing
```

ROIs can also be managed dynamically at runtime via a REST API endpoint served on the Jetson Orin, allowing operators to draw, move, or delete zones from a web dashboard without restarting the pipeline.

STEP 6

Violation Detection & Alert Engine

When a tracked person's bounding box centroid enters a RESTRICTED or BUFFER ROI, the violation detection engine logs the event and dispatches alerts through multiple configured channels simultaneously.

Alert Channels

Channel	Mechanism	Latency	Use Case
On-Screen Overlay	Red bounding box + 'VIOLATION' banner via NvOSD	< 100ms	Operator monitoring screen
MQTT Broker	Publish JSON event to MQTT topic (mosquitto / AWS IoT)	< 200ms	IoT dashboard, cloud integration
HTTP Webhook	POST JSON payload to security management system API	< 500ms	PSIM / VMS integration
Email / SMS Alert	SMTP email or Twilio SMS on critical violations	1–5 sec	Supervisor & security team notification
GPIO Relay Trigger	Activate siren, barrier gate, or warning light via GPIO	< 50ms	Physical access barrier control
Local SQLite Log	Write event record (time, cam ID, person ID, ROI ID)	< 10ms	Audit trail, incident review
RTSP Alert Stream	Annotated video stream routed to NVR / VMS	< 200ms	Video recording with annotations

Alert JSON Payload Example

Sample Violation Event Payload (MQTT / Webhook)

```
{
  "event_type": "ROI_VIOLATION",
  "timestamp": "2026-04-09T14:32:11.452Z",
```

```
  "camera_id": "cam_02_entrance_north",
  "person_track_id": 47,
  "roi_id": "RESTRICTED_ZONE_1",
  "roi_type": "RESTRICTED",
  "bbox": { "x": 342, "y": 198, "w": 68, "h": 195 },
  "centroid": { "x": 376, "y": 296 },
  "confidence": 0.92,
  "frame_id": 18420,
  "stream_resolution": "1920x1080",
  "snapshot_url": "http://jetson-orin-01/snapshots/ev_18420.jpg"
}
```


Deployment Configuration

Software Stack on Jetson Orin

Component	Version / Source	Purpose
JetPack SDK	JetPack 6.x (L4T R36.x)	Base OS, CUDA, cuDNN, TensorRT on Jetson Orin
DeepStream SDK	DeepStream 7.x (NVIDIA NGC)	Core video analytics framework
TensorRT	TensorRT 8.6+ (included in JetPack)	Model optimisation and inference acceleration
TAO Toolkit	TAO 5.x (NVIDIA NGC)	Model fine-tuning on custom datasets
PeopleNet Model	v2.6 (.etlt) — NGC Catalogue	Primary person detection model
CrowdCount Model	DmCount / NVIDIA NGC	Crowd density heatmap generation
OpenCV (CUDA)	4.8+, compiled with CUDA	Pre/post processing, ROI polygon math
GStreamer	1.20+	Pipeline element orchestration
Python / C++ bindings	Python 3.10 + DeepStream Python bindings	Custom plugin development and ROI logic
Mosquitto MQTT Broker	2.x (local) or AWS IoT Core (cloud)	Alert event messaging
InfluxDB + Grafana	InfluxDB 2.x + Grafana 10.x	Time-series metrics storage and dashboard
FastAPI (REST)	FastAPI 0.110+	ROI management REST API on Jetson
SQLite	3.45+	Local event log and violation database
Docker (optional)	Docker + NVIDIA Container Runtime	Containerised deployment for portability

Model Deployment Workflow

- Step 1 — Download PeopleNet .etlt model from NVIDIA NGC catalog
- Step 2 — Convert .etlt to TensorRT engine on Jetson Orin using tao-converter:
  - tao-converter -k <key> -d 3,544,960 -p input_1,1x3x544x960,4x3x544x960,8x3x544x960 -t fp16 -e /models/peoplenet_fp16.engine peoplenet_v2.6.etlt
- Step 3 — Configure DeepStream primary inference config (config_infer_primary_peoplenet.txt)
- Step 4 — Configure NvDCF tracker with appropriate parameters for crowd density
- Step 5 — Define ROI polygons in config_nvdsanalytics.txt for each camera stream
- Step 6 — Configure alert sinks: MQTT broker address, webhook URLs, GPIO pins
- Step 7 — Launch DeepStream application: deepstream-app -c deepstream_crowd_config.txt
- Step 8 — Verify annotated RTSP output stream on monitoring dashboard

ROI Management Interface

A lightweight web-based ROI Management Dashboard is deployed alongside the DeepStream pipeline on the Jetson Orin. Operators can use this interface to draw, edit, and delete ROI zones directly on a live camera snapshot without editing configuration files.

Dashboard Features

- Live Camera Snapshot Viewer — Display current frame from each camera for ROI placement
- ROI Drawing Tool — Click-to-draw polygonal zones directly on the camera image
- ROI Type Assignment — Set zone type: Restricted / Buffer / Crowd Limit / Direction / Dwell
- Per-Zone Settings — Configure crowd threshold, dwell time limit, active schedule, and alert severity
- Real-Time Violation Feed — Live log of detected violations with thumbnails
- Crowd Density Heatmap View — Overlay density map on camera stream in real time
- Person Count per Zone — Live occupancy count displayed per zone
- Incident Replay — Access stored violation snapshots and 30-second event video clips
- REST API — All dashboard actions also exposed via REST API for integration with VMS/PSIM

Dashboard Tech Stack
Frontend: React.js + Fabric.js (canvas-based polygon drawing) + Leaflet.js (zone visualisation)
Backend: FastAPI (Python) running on Jetson Orin, serving REST endpoints for ROI CRUD operations
Communication: WebSocket for live violation event feed; REST for ROI config updates
Video: RTSP-to-HLS transcoder (FFmpeg) for in-browser live camera preview
Persistence: SQLite for ROI zone definitions; InfluxDB for time-series crowd metrics

Tools, Components & Budget Estimate

The following table covers all software tools, models, and infrastructure components required for the system. Hardware costs are one-time capital expenditure (CapEx); software costs are ongoing operational expenditure (OpEx).

Software & Services (OpEx)

Tool / Service	Purpose	Licence / Pricing	Est. Monthly Cost
NVIDIA DeepStream SDK	Core video analytics pipeline	Free (NVIDIA SDK)	$0
NVIDIA PeopleNet v2.6	Person detection model	Free (NVIDIA NGC)	$0
NVIDIA CrowdCount Model	Crowd density estimation	Free (NVIDIA NGC)	$0
NVIDIA TAO Toolkit	Model fine-tuning on custom data	Free (NVIDIA NGC)	$0
JetPack SDK 6.x	Jetson OS + CUDA/TensorRT/cuDNN	Free	$0
OpenCV (CUDA build)	Image processing, polygon math	Open-source (Apache 2.0)	$0
Mosquitto MQTT Broker	Local alert event messaging	Open-source (EPL 2.0)	$0
InfluxDB 2.x	Time-series crowd metrics storage	OSS: Free / Cloud: $25+/mo	$0–$25
Grafana 10.x	Metrics dashboard & visualisation	OSS: Free / Cloud: $10+/mo	$0–$10
FastAPI + React.js	ROI management web dashboard	Open-source	$0
AWS IoT Core (optional)	Cloud MQTT for multi-site alerting	$0.08/100K messages	$5–$30
Twilio SMS (optional)	SMS alerts to security personnel	$0.0079/SMS	$5–$20
Docker + NGC Container	Containerised deployment	Free	$0
GitHub / GitLab	Source code management	Free / $4 per user/mo	$0–$20

Hardware (CapEx — One-Time Cost per Site)

Component	Specification	Qty (Typical)	Unit Cost	Total Cost
Jetson AGX Orin 32GB	200 TOPS, 32GB RAM, NVMe slot	1	$999	$999
NVMe SSD 512GB	Samsung 980 Pro M.2	1	$70	$70
IP Camera (1080p)	H.265 RTSP, PoE, 2.8mm fixed lens	4–8	$120	$480–$960

PoE Switch (8-port)	802.3af, 65W total PoE budget	1	$80	$80
Active Cooling Kit	Official Jetson Orin AGX heatsink	1	$35	$35
GPIO Relay Module	4-channel 5V relay, DIN mountable	1	$15	$15
UPS (12V, 4Ah)	2–4hr runtime, automatic cut-over	1	$120	$120
IP65 Enclosure	Outdoor-rated, lockable steel box	1	$90	$90
CAT6 Cabling + PoE	Per-camera run, conduit	4–8	$30	$120–$240
Warning Siren + Light	12V strobe + 110dB horn	1–2	$25	$25–$50

Budget Summary

Budget Category	Details	Estimated Cost
Hardware CapEx (Small Site)	Orin AGX 32GB + 4 cameras + accessories	~$2,000 – $2,500
Hardware CapEx (Medium Site)	Orin AGX 32GB + 8 cameras + accessories	~$2,600 – $3,400
Hardware CapEx (Large Site)	2× Orin AGX 64GB + 16 cameras + accessories	~$6,000 – $8,000
Software (All open-source)	DeepStream, PeopleNet, OpenCV, Mosquitto, etc.	$0 (fully open-source)
Cloud Services (optional)	AWS IoT Core + Twilio SMS (moderate usage)	~$10 – $50/month
Dashboard Hosting	Self-hosted on Jetson Orin	$0
Integration / Dev Labour	Initial setup + custom plugin dev (one-time)	~$5,000 – $15,000
Annual Maintenance	Firmware updates, model refresh, support	~$1,000 – $3,000/yr

Model Fine-Tuning with NVIDIA TAO Toolkit

While PeopleNet performs well out-of-the-box, fine-tuning on site-specific data (e.g., top-down camera angles, low-light conditions, specific clothing or uniforms) significantly improves detection accuracy. NVIDIA TAO Toolkit enables fine-tuning without requiring deep ML expertise.

- Collect 500–2,000 labelled images from the deployment site cameras
- Annotate using NVIDIA CVAT (open-source) or Roboflow (SaaS, free tier available)
- Run TAO fine-tuning on a cloud GPU instance (AWS/GCP/Azure) or locally on the Jetson
- Export fine-tuned model as .etlt and re-convert to TensorRT engine for deployment
- Expected accuracy improvement: +5–15% mAP in site-specific conditions vs base model

TAO Fine-Tuning Resources
TAO Toolkit Documentation: https://docs.nvidia.com/tao/tao-toolkit/

PeopleNet Fine-Tuning Guide: Available on NVIDIA Developer Zone
Dataset annotation tool: NVIDIA CVAT — https://cvat.ai (open-source, self-hostable)
Recommended training GPU: NVIDIA A10G (AWS g5 instance) — ~$1.00/hr for ~2–4hr training run
Estimated fine-tuning cost: $2–$8 per training run on cloud GPU

Performance Benchmarks

The following benchmarks are based on NVIDIA published figures and community testing for DeepStream with PeopleNet on Jetson Orin hardware:

Metric	Jetson Orin NX 16GB	Jetson AGX Orin 32GB	Jetson AGX Orin 64GB
Max concurrent 1080p streams	4–6 streams	12–16 streams	16–20 streams
PeopleNet inference FPS (per stream)	~25–30 FPS	~30 FPS	~30 FPS
End-to-end pipeline latency	~80–120ms	~50–80ms	~40–70ms
ROI violation detection latency	< 150ms	< 100ms	< 80ms
Power consumption (pipeline active)	~15–20W	~25–40W	~40–60W
People detection accuracy (mAP50)	~84% (base)	~84% (base)	~84% (base)
GPU utilisation (full load)	~70–85%	~60–75%	~50–65%

Risks & Mitigations

Risk	Severity	Mitigation
False positives — benign persons flagged as violations	High	Tune confidence thresholds; add 2-frame persistence check before triggering alert; include operator confirmation for physical barriers.
Occlusion causing missed detections in dense crowds	High	Use multiple overlapping camera angles; enable Re-ID model; tune tracker max-age settings for dense scene.
Thermal throttling of Jetson under sustained load	Medium	Install active cooling kit; monitor GPU temp via jtop; set max performance mode via jetson_clocks.
Network failure cutting off MQTT/webhook alerts	Medium	Primary alert via GPIO relay (no network needed); implement local queue with retry for MQTT messages.
Privacy / GDPR concerns with persistent person tracking	High	Do not transmit raw video off-device; use track IDs not biometrics; configure data retention policy (auto-purge events > 30 days).
Camera angle changes invalidating stored ROI coordinates	Low	Tie ROI configs to camera serial ID; detect frame homography shift; alert admin if camera is moved.

Model performance degradation in low-light or night conditions	Medium	Use IR-capable cameras; fine-tune model on night-time site images; lower confidence threshold for night mode.

Implementation Roadmap

Phase 1 — Proof of Concept (Weeks 1–3)

- Procure Jetson AGX Orin 32GB dev kit and 2 test IP cameras
- Flash JetPack 6.x, install DeepStream SDK and dependencies
- Download and convert PeopleNet .etlt to TensorRT engine
- Run DeepStream people-detection reference app on test camera feed
- Define 2–3 test ROI zones using nvdsanalytics config
- Verify violation detection output on single stream at 30 FPS

Phase 2 — Core Pipeline Integration (Weeks 4–7)

- Integrate multi-camera support (4–8 streams via NvStreamMux)
- Build custom ROI violation GStreamer plugin in Python
- Implement MQTT alert publishing and GPIO relay trigger
- Deploy FastAPI ROI management REST API on Jetson
- Integrate CrowdCount model for density heatmap overlay
- Build basic React.js dashboard for ROI drawing and live event feed

Phase 3 — Production Hardening (Weeks 8–12)

- Fine-tune PeopleNet on site-specific data using TAO Toolkit
- Integrate InfluxDB + Grafana for crowd analytics dashboard
- Implement alert escalation logic (email/SMS on sustained violations)
- Add incident replay and snapshot export to dashboard
- Deploy Docker container for reproducible rollouts across sites
- Conduct load testing (16 streams, sustained 8-hour run) and thermal validation
- Document deployment runbook and train security operations team

Conclusion & Recommended Next Steps

The proposed NVIDIA DeepStream Crowd Detection & ROI Enforcement System provides a fully edge-deployed, real-time solution for intelligent crowd management and access restriction. Leveraging NVIDIA's pre-trained PeopleNet and CrowdCount models on the Jetson Orin hardware platform ensures production-grade inference performance with minimal recurring cost — all primary software components are open-source and free.

The system's ROI-based enforcement architecture is flexible enough to cover a wide range of scenarios — from industrial no-entry zones to stadium crowd management — and the interactive ROI dashboard eliminates the need for technical expertise to manage zones post-deployment.

Recommended immediate next steps:

- Approve hardware procurement: Jetson AGX Orin 32GB dev kit (~$999) + 2 test cameras
- Assign 1 ML engineer + 1 embedded software engineer to Phase 1
- Register on NVIDIA NGC and download PeopleNet v2.6 model
- Identify target deployment site and survey camera mounting positions
- Define initial set of ROI zones in coordination with security operations team
- Schedule Phase 1 kick-off for a 3-week proof-of-concept sprint

All cost estimates are approximate and based on publicly available pricing as of April 2026. Hardware specifications subject to change.